

WEEK 4 - TUESDAY: PRACTICE QUESTIONS

QUESTION 1: Pizza Distribution Problem

A class is organizing a pizza party with 32 students. They order 5 pizzas, and each pizza has 8 slices. Each slice costs Rs. 120.

TASKS:

1. Calculate the total number of slices available
2. Calculate how many slices each student will get (equal distribution)
3. Calculate how many slices will be left over
4. Calculate the total cost of the pizzas
5. Display all the above information

TOPICS: Arithmetic operators (*, /, %)

EXPECTED OUTPUT:

Total slices: 40

Each student gets: 1 slice(s)

Leftover slices: 8

Total cost: Rs. 4800

QUESTION 2: Student Exam Eligibility Check

A student wants to determine if they are eligible to sit for the final exam.

The eligibility criteria are:

- Attendance must be at least 75%
- Assignment score must be greater than 50
- No outstanding fees

A student has:

- Attendance: 80%
- Assignment Score: 47
- Outstanding Fees: No

TASKS:

1. Check if attendance meets the requirement ($\geq 75\%$)
2. Check if assignment score meets the requirement (> 50)
3. Check if there are no outstanding fees
4. Determine overall eligibility (all three conditions must be true)
5. Display all findings

TOPICS: Logical operators (&&), Comparison operators ($\geq$, $>$), Negation operator (!)

EXPECTED OUTPUT:

Attendance OK: true

Assignment OK: false

No outstanding fees: true

Eligible for exam: false

NOTE: The student is NOT eligible because the assignment score (47) is NOT greater than 50, even though attendance is sufficient and there are no fees.

QUESTION 3: Savings Tracker with Assignment Operators

Track changes to your savings account through multiple transactions using assignment operators.

SCENARIO:

- Initial savings: Rs. 5000
- You earn Rs. 1200 from tutoring and add it to your savings
- You spend 1/3 of your total savings on books
- You sell some gadgets and earn enough to double your current savings

TASKS:

1. Add the tutoring income to savings and display the result
2. Divide your savings by 3 (after buying books) and display the result
3. Multiply your savings by 2 (from selling gadgets) and display the result

*TOPICS: Assignment operators (+=, /=, *=)*

EXPECTED OUTPUT:

After tutoring income: 6200.0

After buying books: 2066.6666666666665

After selling gadgets: 4133.333333333333

BREAKDOWN:

Step 1: $5000 + 1200 = 6200$

Step 2: $6200 / 3 = 2066.67$ (approximately)

Step 3: $2066.67 * 2 = 4133.33$ (approximately)

QUESTION 4: Food Delivery Fee and Discount Calculation

A food delivery service has policies for applying delivery fees and discounts based on order amount, distance, and membership status.

POLICIES:

- Delivery Fee: Rs. 60 is charged if the order is less than Rs. 500 OR the delivery distance exceeds 10 km. Otherwise, no fee.
- Discount: Premium members get Rs. 100 discount if the order is greater than Rs. 800. Otherwise, no discount.

CASE:

- Order Total: Rs. 650
- Distance: 12 km
- Premium Member: Yes

TASKS:

1. Determine if delivery fee applies (order < 500 OR distance > 10)
2. Determine if discount applies (order > 800 AND premium member)
3. Calculate and display the final amount (order + fee - discount)

TOPICS: Logical operators (||, &&), Ternary operator (? :), Comparison operators

EXPECTED OUTPUT:

Delivery fee applies: true

Discount applies: false

Final amount: Rs. 710.0

BREAKDOWN:

- Fee applies: YES, because distance (12 km) > 10 km
- Discount applies: NO, because order (650) is NOT > 800
- Final amount: $650 + 60 - 0 = 710$

